

20 April 2026

String Manipulation

Replace ← only single instance

Replace All ← all instances/patterns/regular expressions

Split/Split → divide a string

Replace All

String s = "Hello123 BUBT09" [a-z] [A-Z] [aeiouAEIOU]

s = s.replaceAll("[0-9]", "#");

Split

কিছু string বা substring এ ভাগ করতে

String s = "Hello BUBT"

String[] arr = s.split(" ");

for (String a : arr)

System.out.println(a);

split(string, optional limit)

study all materials from

☐ PDF & include split

& replace All

File Read & Write (Basic File Manipulation)

import java.io.*;

public class Main {

public static void main(String[] args) {

try {

FileWriter w = new FileWriter("E:\\dir\\in.txt");

w.write("Hello, BUBT");

w.close();

} catch (IOException e) {

System.out.println(e);

}

Example: 10 number user input & write to file

Life is less complicated when you have a cat.

```
import java.util.*;
import java.io.*;
```

Shayan

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(); int
        for(int i=0; i<10; i++) {
        try {
            FileWriter w = new FileWriter("C:\\out.txt");
            for(int i=0; i<10; i++) {
                int x = sc.nextInt();
                String xs = x + "\n";
                w.write(xs);
            }
            w.close();
        } catch (IOException e) {
            System.out.println(e);
        }
    }
}
```

#File Reading (File or FileReader class)

```
File r = new File("E:\\input.txt");
Scanner sc = new Scanner(r);
while(sc.hasNext()) {
    int a = sc.hasNextInt();
    System.out.println(a);
}
sc.close();
```

Wrap all of it in the try-catch block

PDF এ অনেক কিছু আছে just read & write দেখানো হবে।

```
import java.util.*;
import java.io.*;

public class Main {
    public static void for main(String[] args) {
        try { // Read input file
            File r = new File("C:\\input.txt");
            Scanner sc = new Scanner(r);
            String[] lines; int count = 0;
            while (sc.hasNext()) {
                lines.push(sc.nextLine());
            }
            sc.close();
            String myName = lines[0]; // assuming first line
                                   // contains the name

            String[] partitions = myName.split(" ");
            partitions[0] = partitions[0].toUpperCase();
            partitions[1] = partitions[1].toUpperCase();

            /*
            * for(int i=0; i<partitions.length; i++) {
            *     partitions[i] = partitions[i].toUpperCase();
            * }
            */

            String NewName = partitions.join(" ");

            /* String String NewName = " ";
            * for (int i=0; i<partitions.length; i++) {
            *     NewName += partitions[i];
            * }
            */
        }
    }
}
```


// write to file

Shayan

```
FileWriter w = new FileWriter("C:\\out.txt");  
w.write(NewName);  
w.close();
```

```
} catch (IOException e) {  
    System.out.println(e);  
}
```

Read all numbers from a file
then write odd numbers to file1.txt,
 >> even numbers to file2.txt,
 >> prime numbers to file3.txt

→ string operations সুনির্দিষ্ট combined হিসাবে আসবে

① Exception Handling (2 Questions)

- ↳ Built-in (Arithmetic, NullPointerException, OutOfBounds, IllegalArgument)
- ↳ User Defined

② String Handling (Combined Operations) <1> + File (?)

③ Java Collection Framework (2 Questions)

- ↳ Array, Linked List, Stack, HashSet, HashMap,

- ↳ Treeset, TreeMap, PriorityQueue

- ↳ Comparable & Sort

- ↳ accessing "isEmpty()

- ↳ collection sort X doesn't work

④ Multithreading (Thread Life Cycle)

⑤ Generics (Method/Class/Interface) (Easy Questions)

⑥ File Handling (1 Questions)